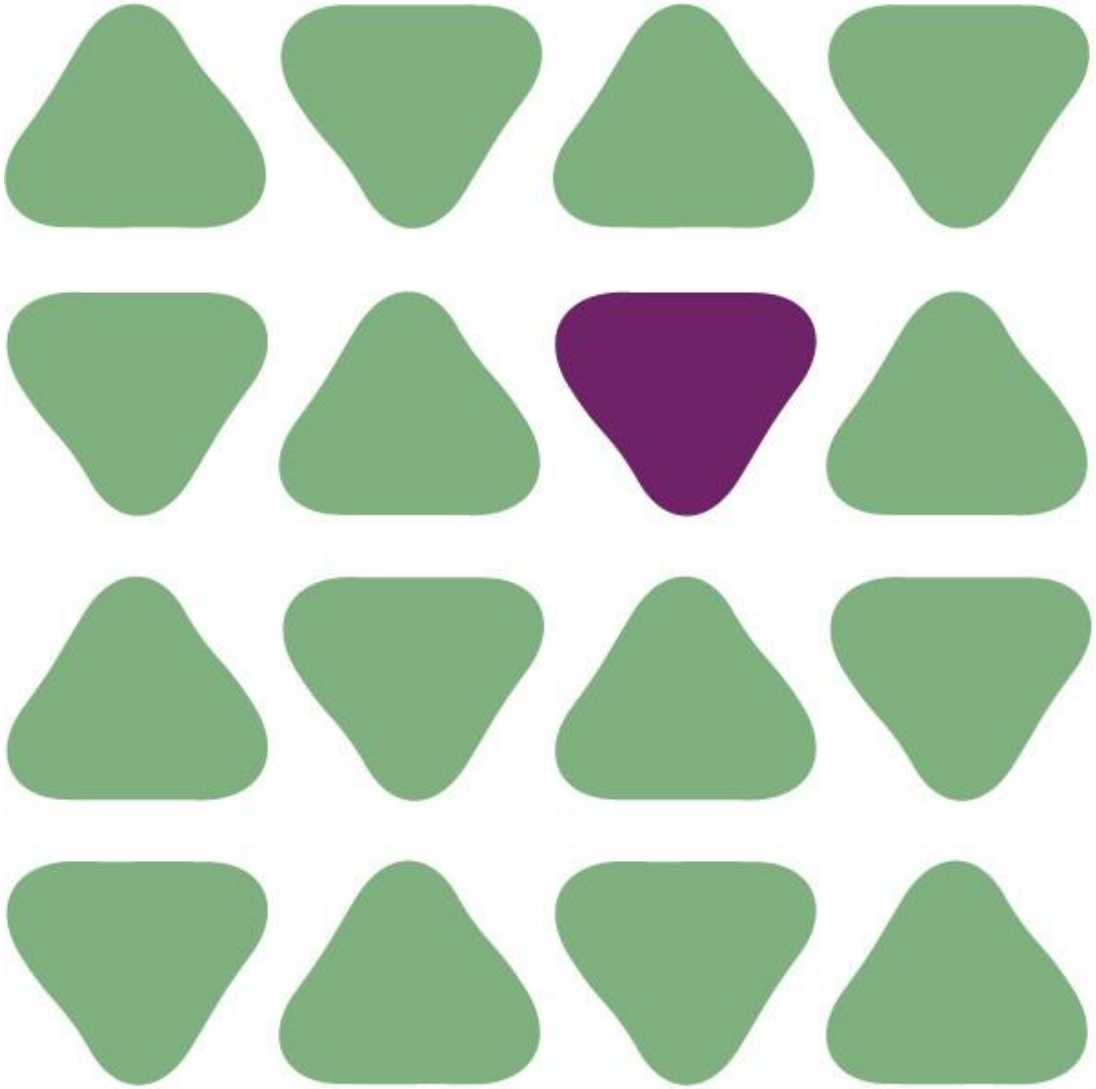
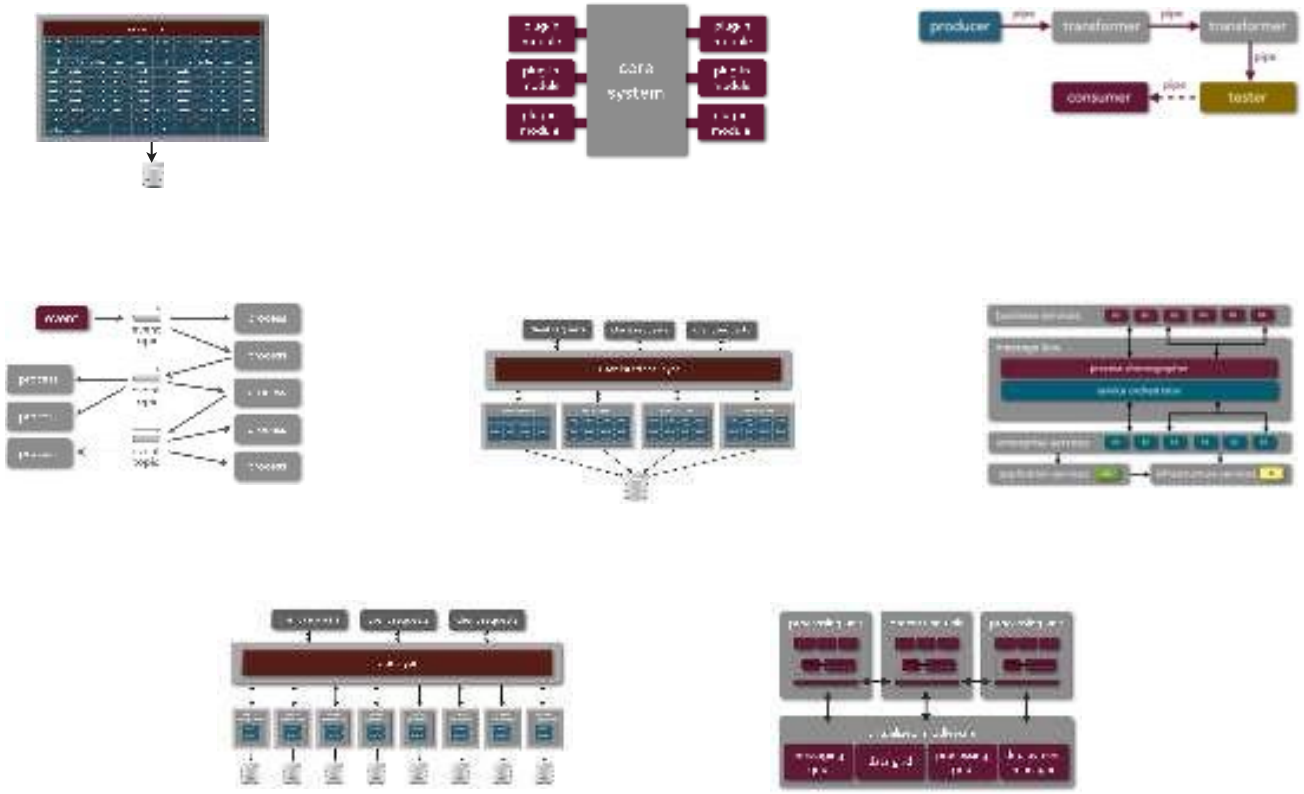
Architecture Patterns



architecture patterns help define the basic characteristics and behavior of the application



architecture pattern classification

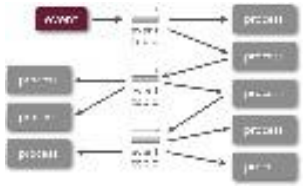
monolithic

layered

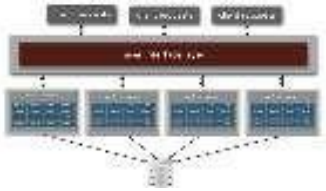
microkernel

pipeline

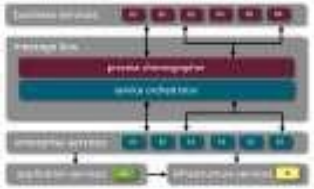
distributed



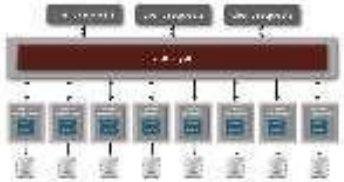
event-driven



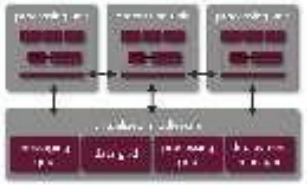
service-based



service-oriented



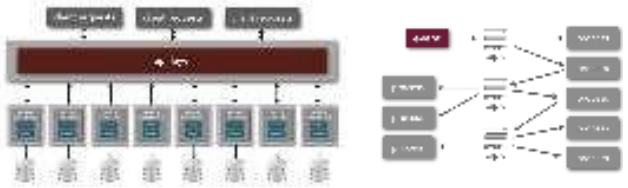
microservices



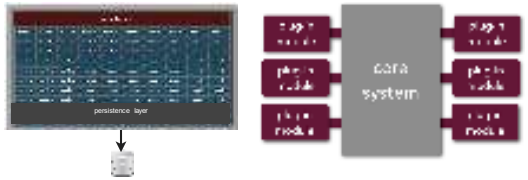
space-based

architecture pattern hybrids

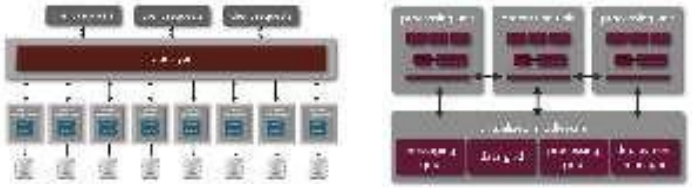
event-driven layered



event-driven microservices

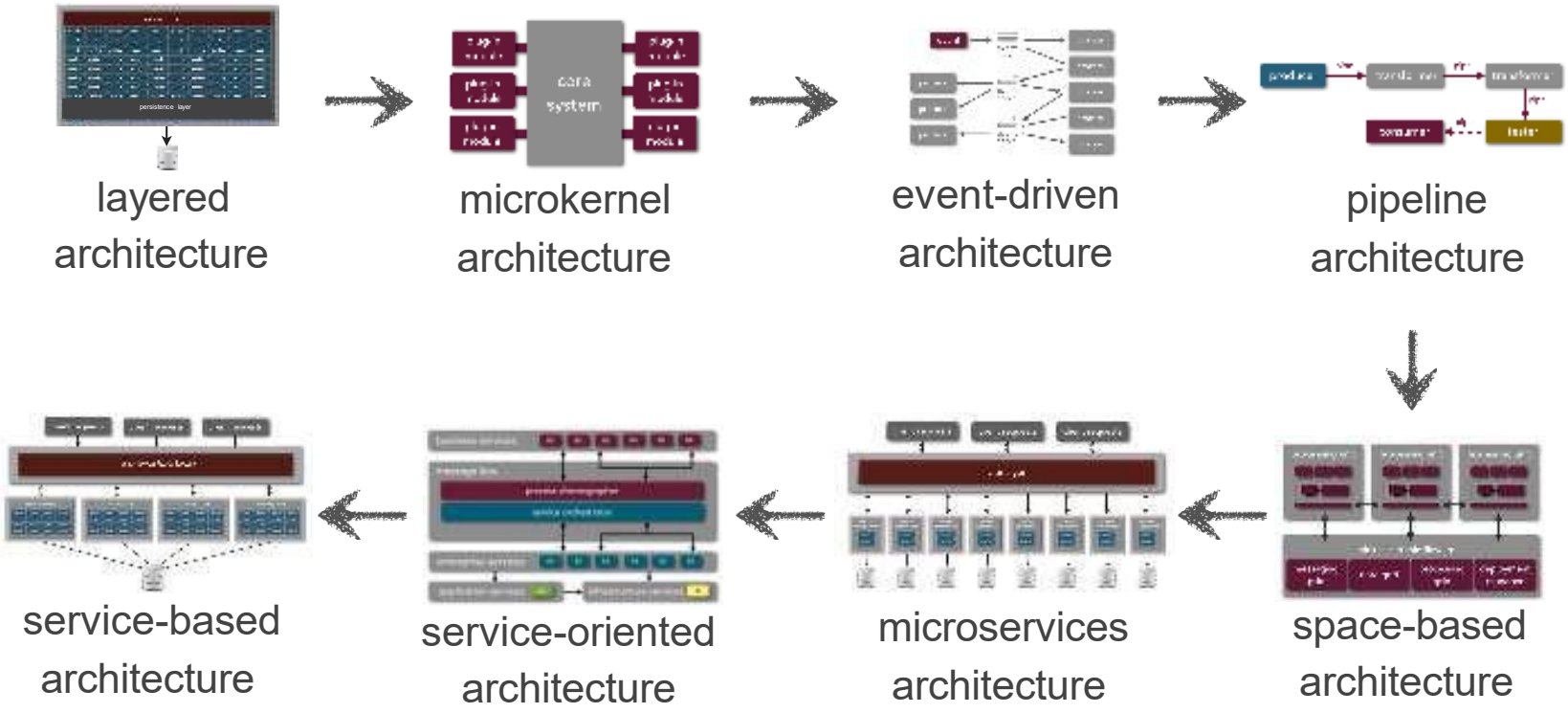


layered microkernel

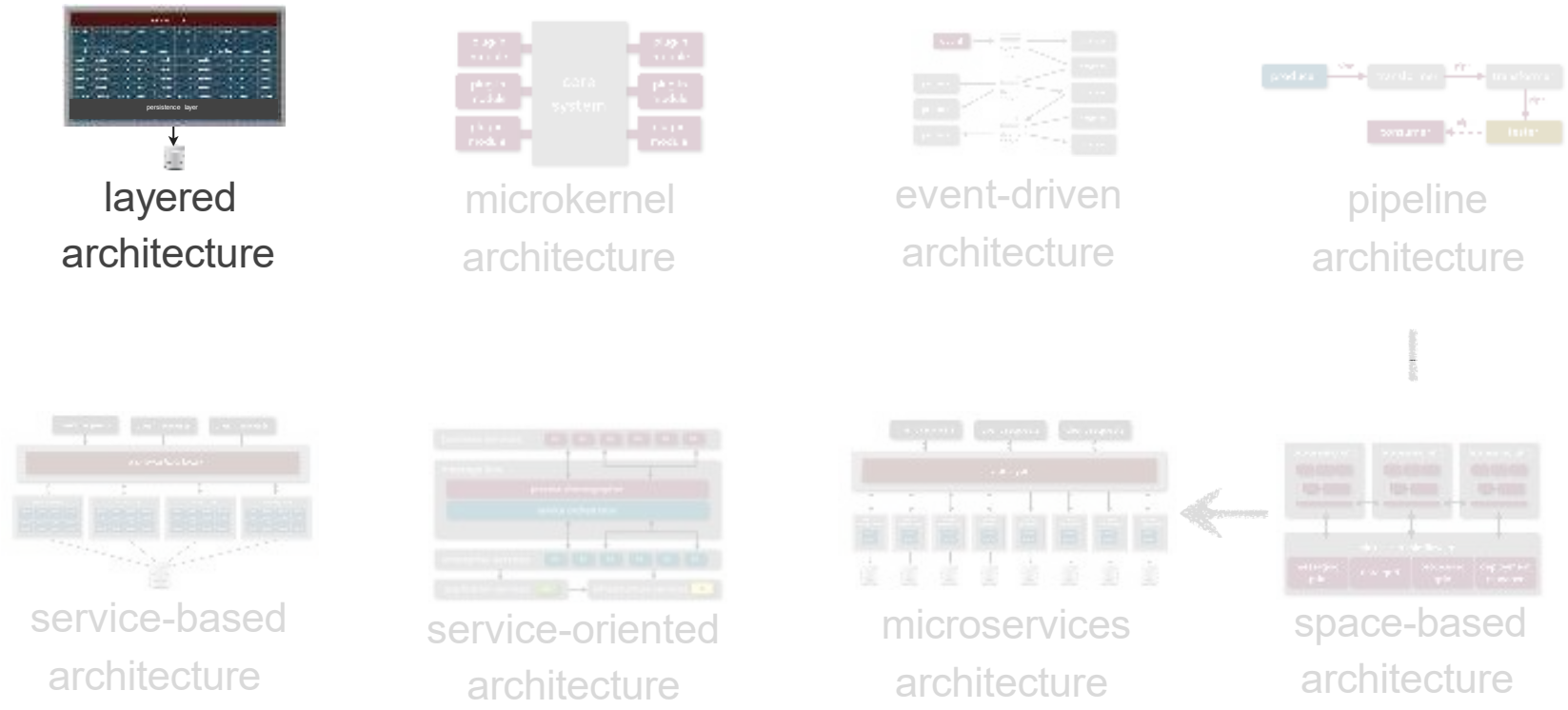


space-based microservices

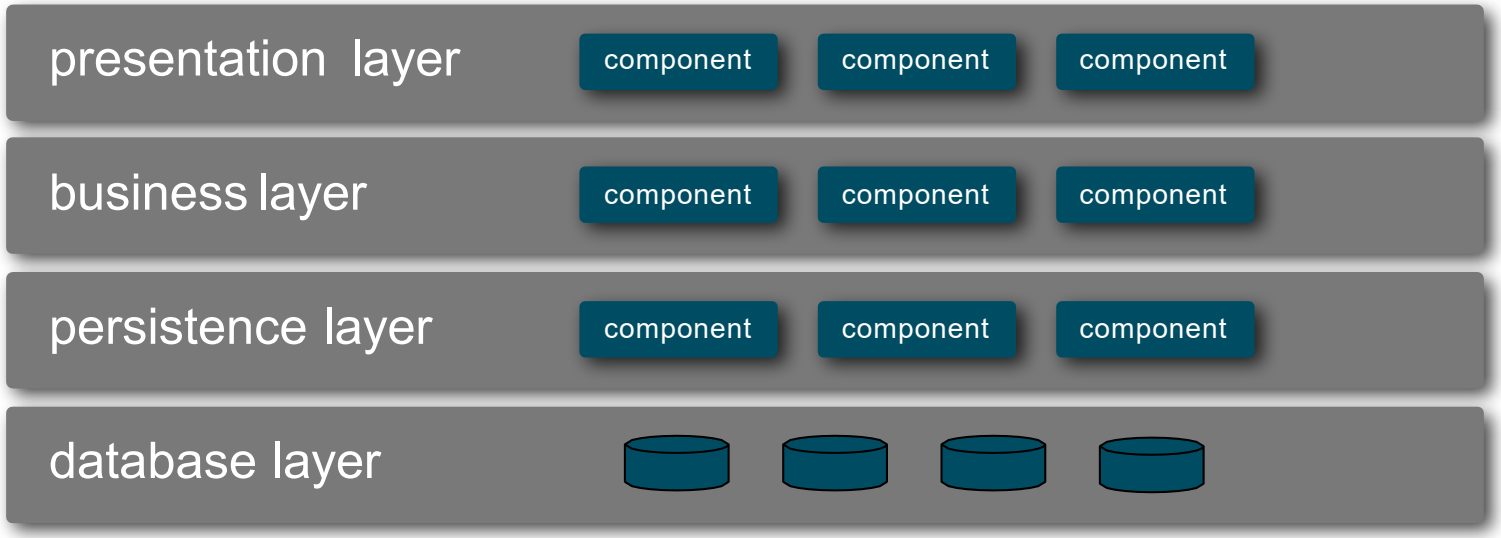
architecture pattern roadmap



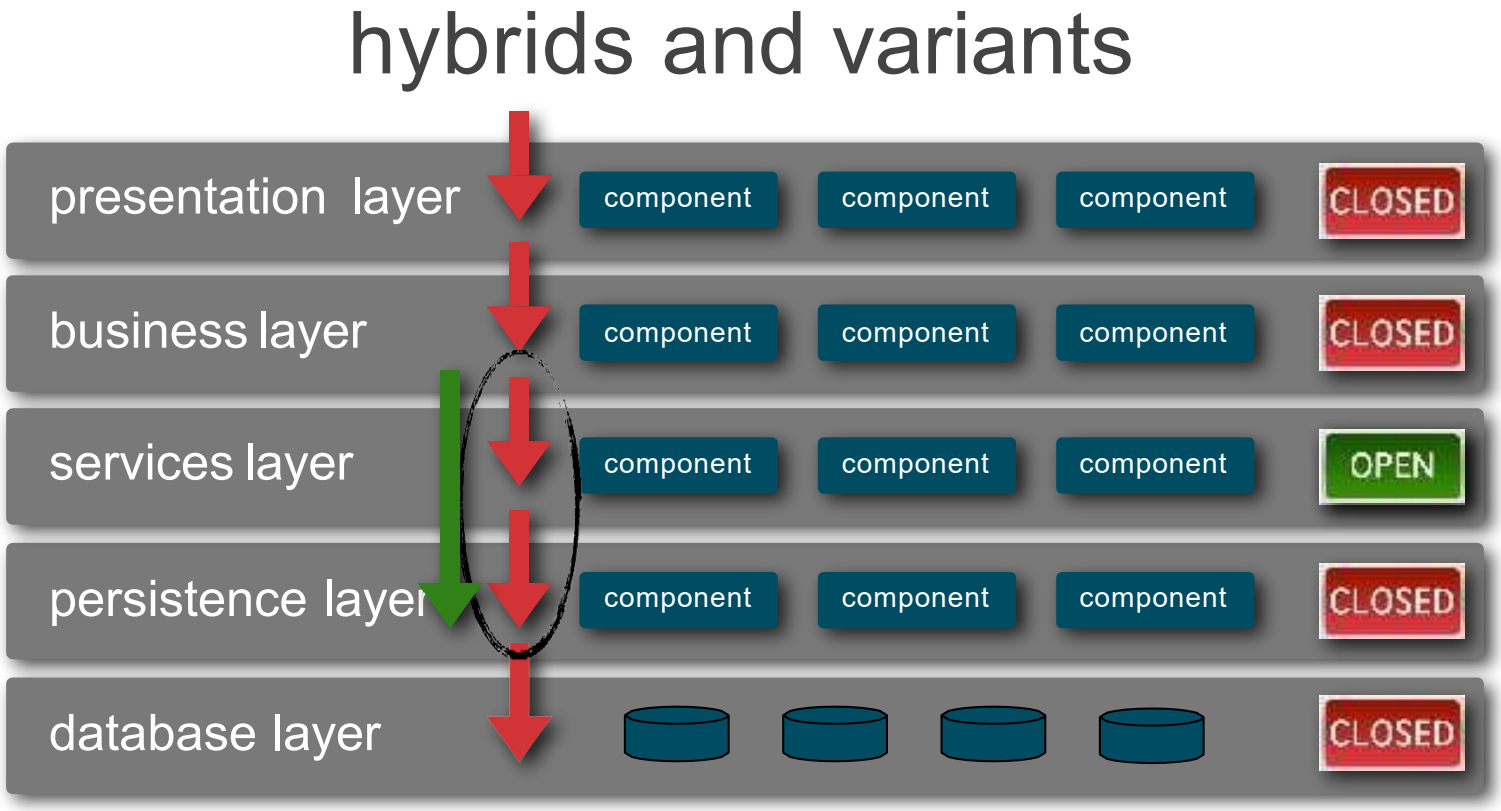
architecture pattern roadmap



layered architecture



layered architecture



architecture pattern roadmap

layered architecture

microkernel architecture

event-driven architecture

pipeline architecture

service-based architecture

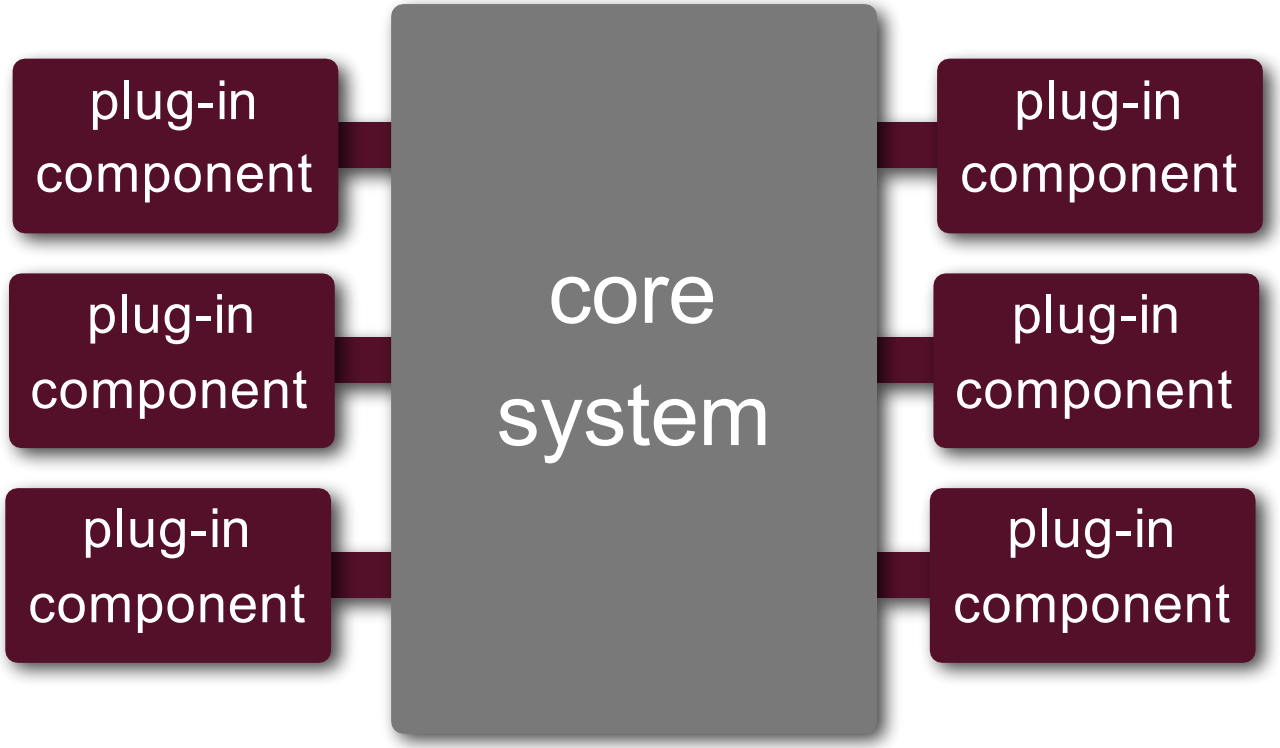
service-oriented architecture

microservices architecture

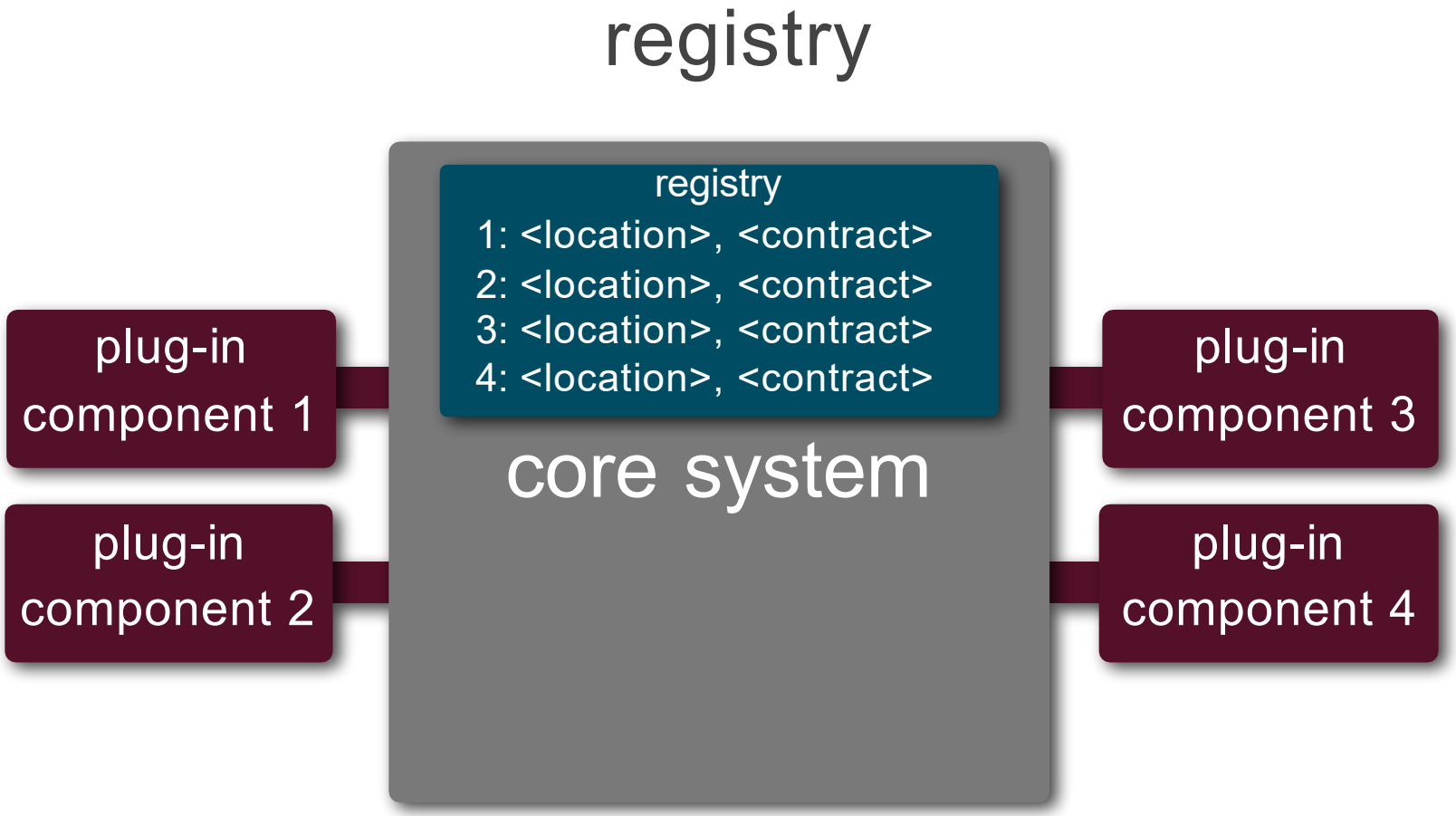
space-based architecture

microkernel architecture

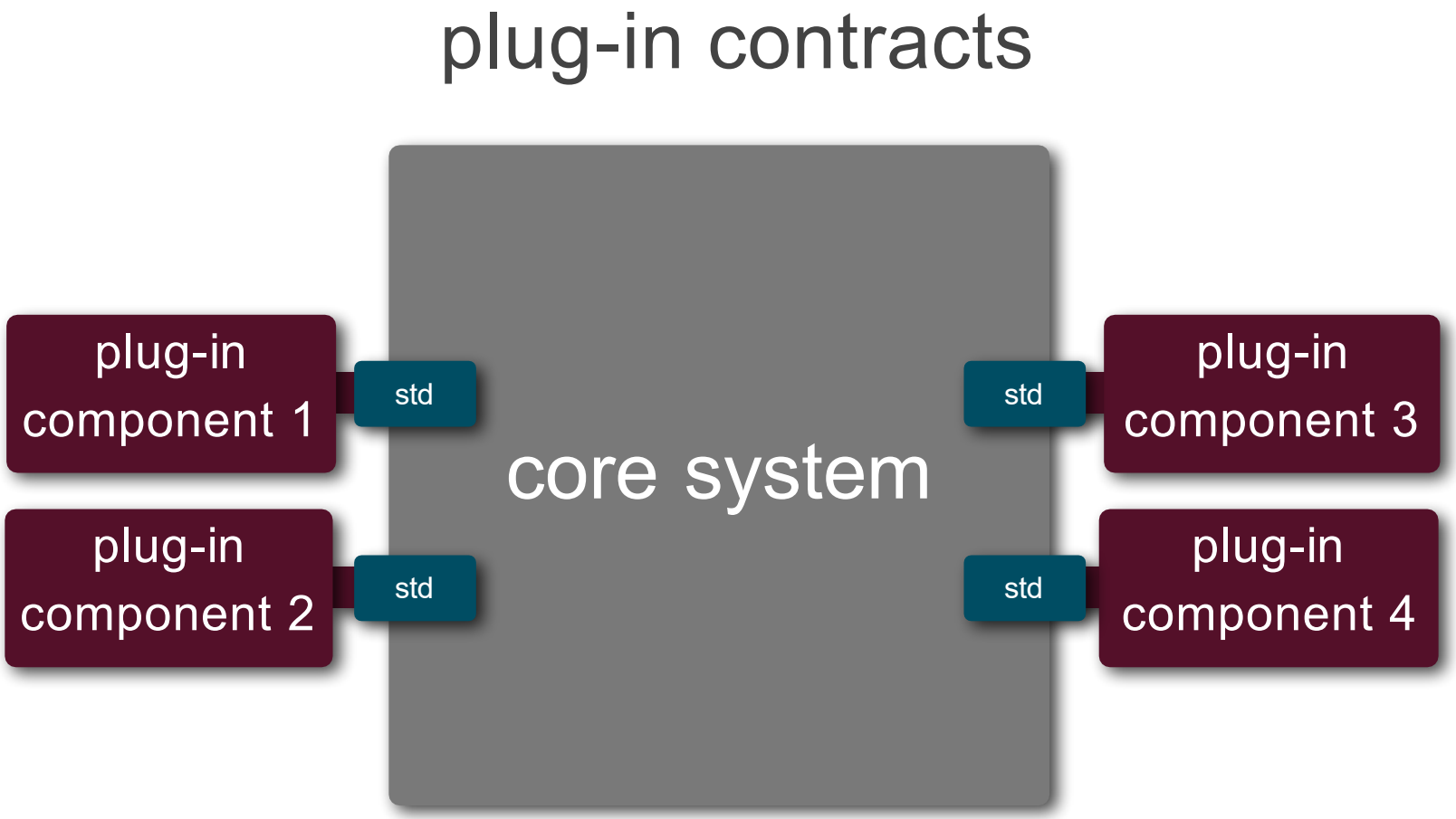
(a.k.a. plug-in architecture pattern)



microkernel architecture



microkernel architecture



architecture pattern roadmap

layered architecture

microkernel architecture

event-driven architecture

pipeline architecture

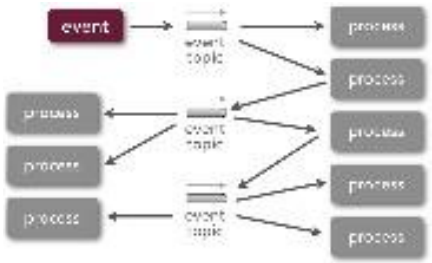
service-based architecture

service-oriented architecture

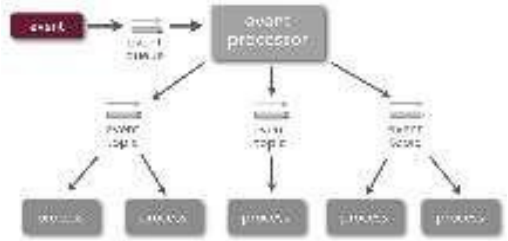
microservices architecture

space-based architecture

event-driven architecture

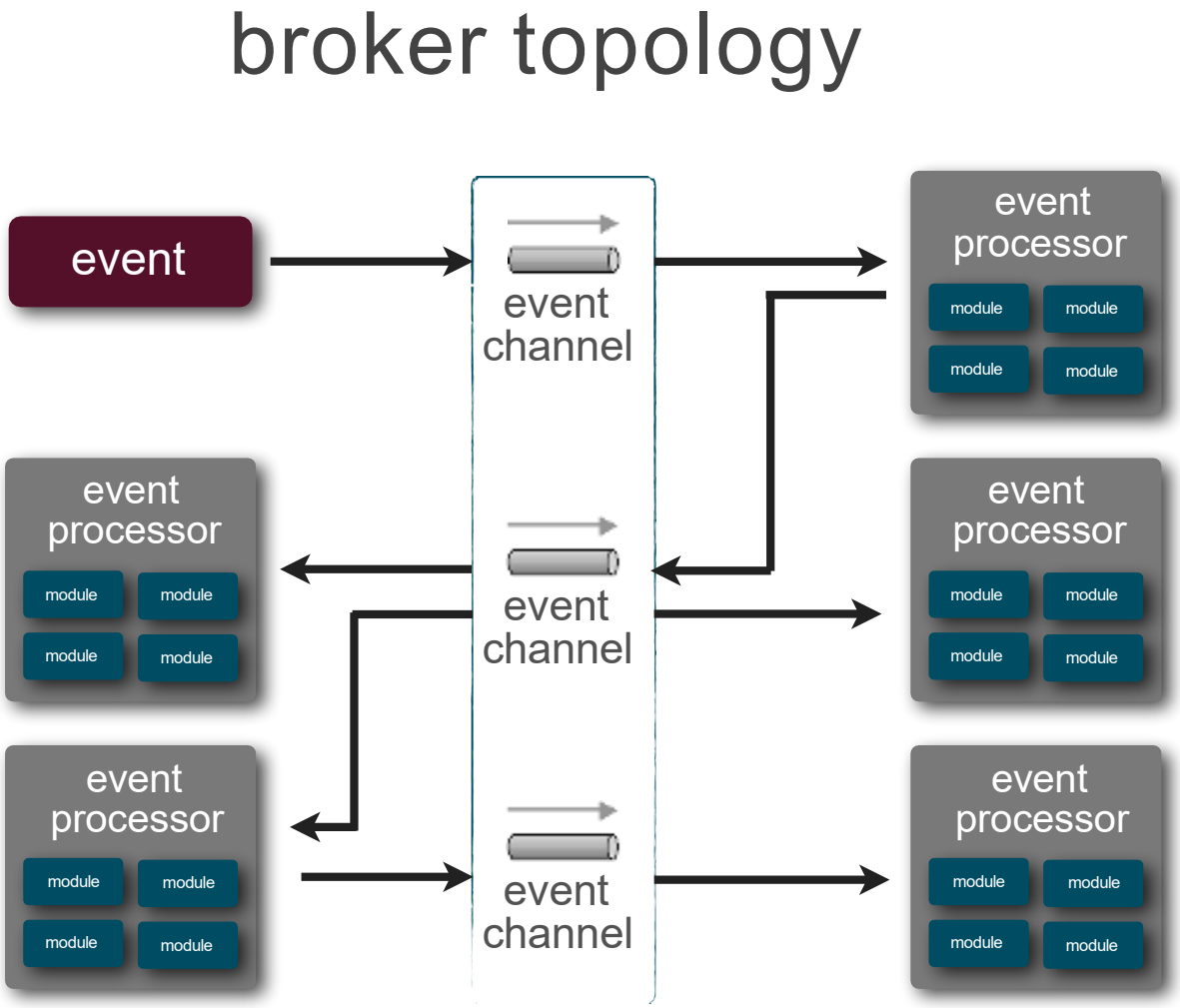


broker topology

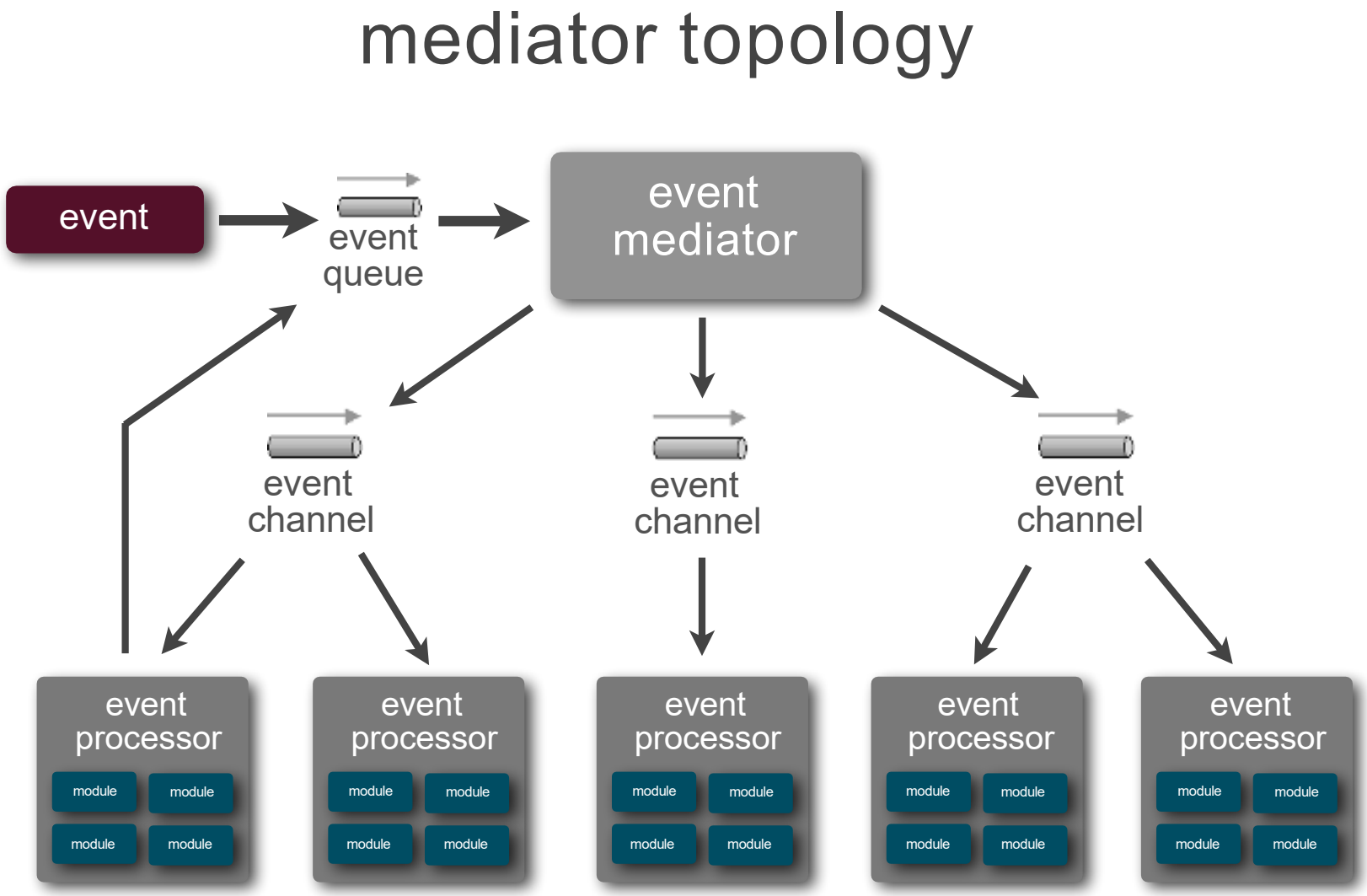


mediator topology

event-driven architecture



event-driven architecture



architecture pattern roadmap

layered architecture

microkernel architecture

event-driven architecture

pipeline architecture

service-based architecture

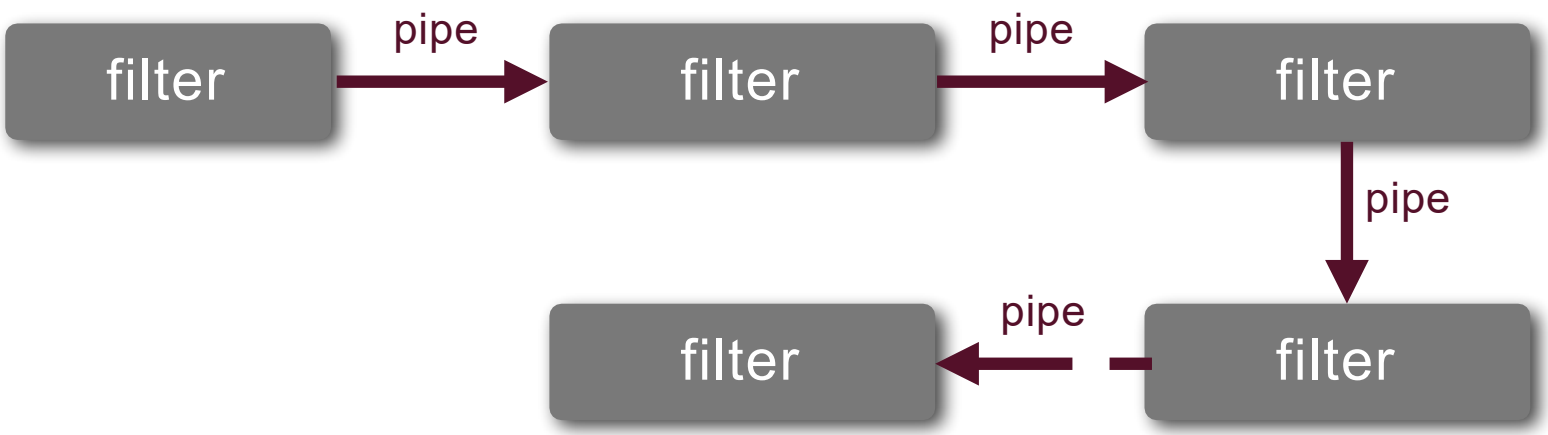
service-oriented architecture

microservices architecture

space-based architecture

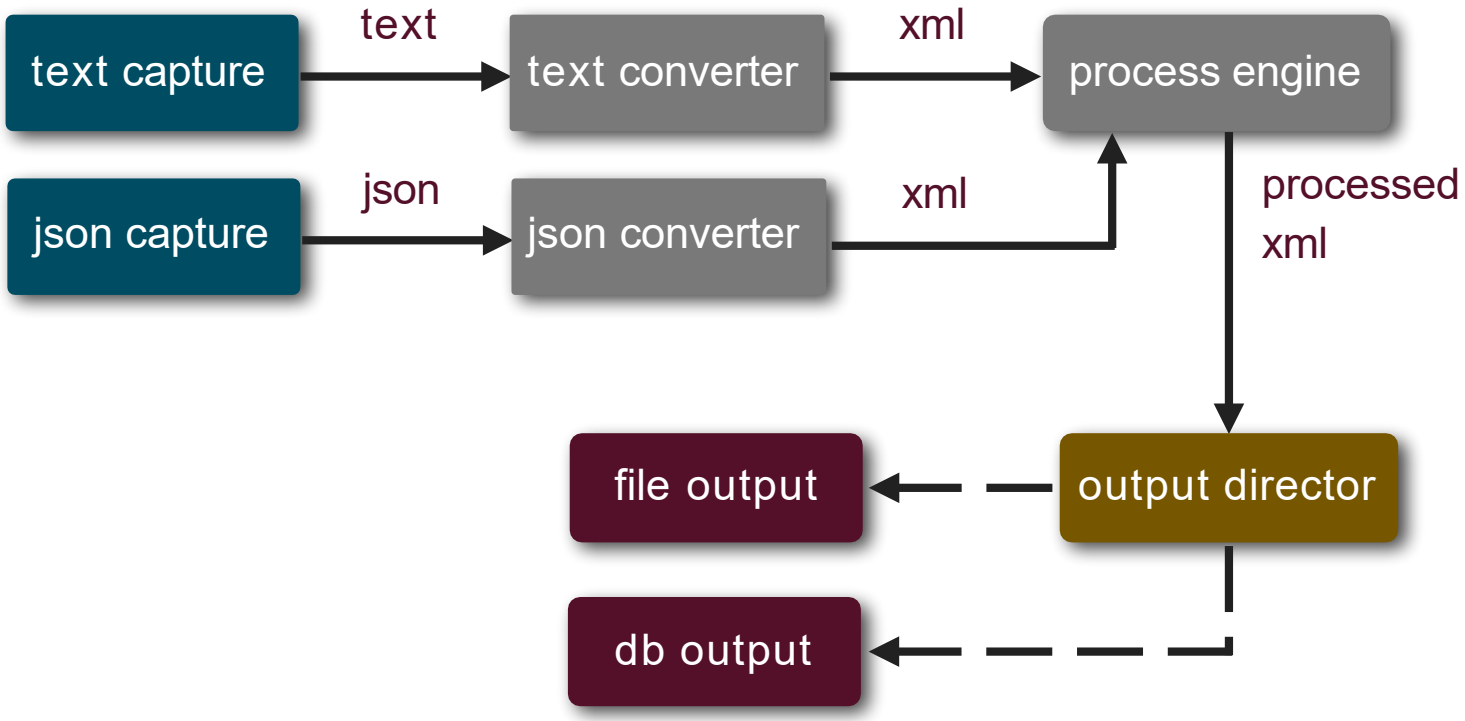
pipeline architecture

(a.k.a. pipe and filter architecture)



pipeline architecture

example: capture data in multiple formats, process the data, and send to multiple outputs



architecture pattern roadmap

layered architecture

microkernel architecture

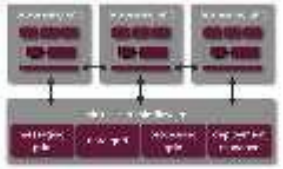
event-driven architecture

pipeline architecture

service-based architecture

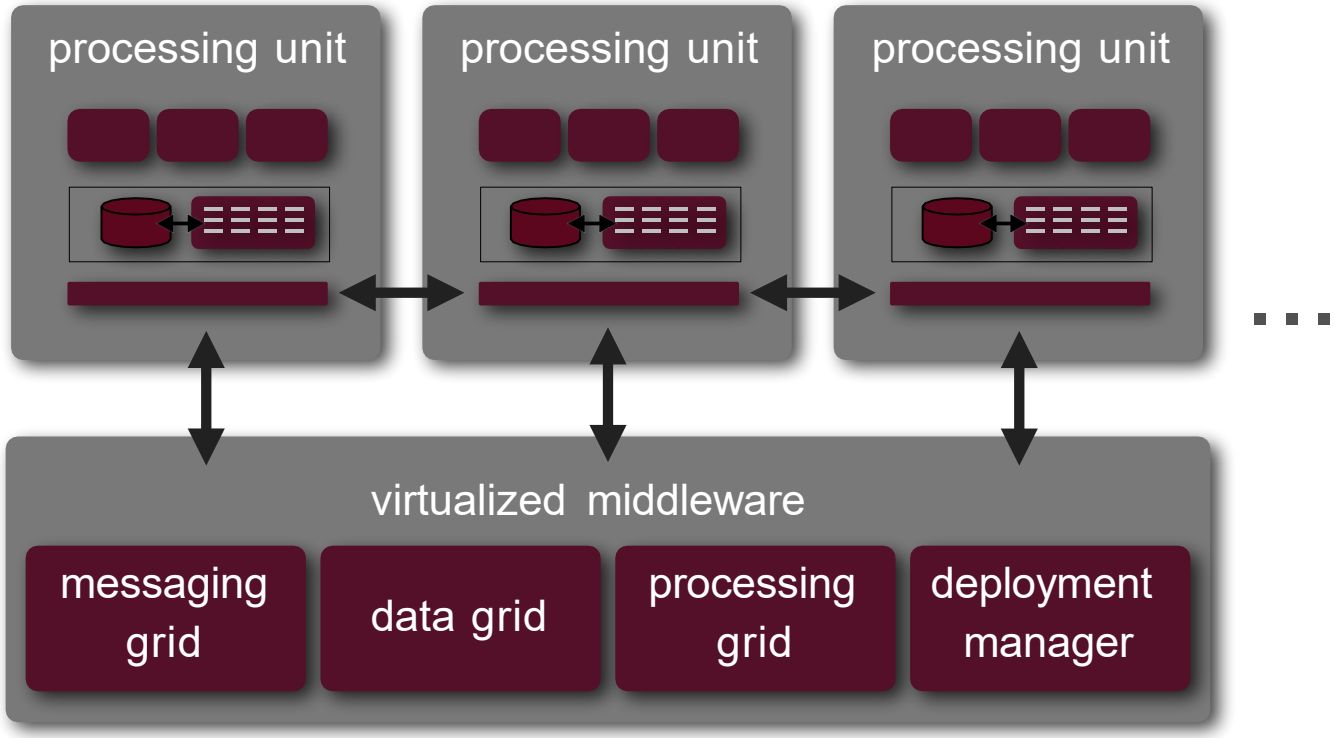
service-oriented architecture

microservices architecture



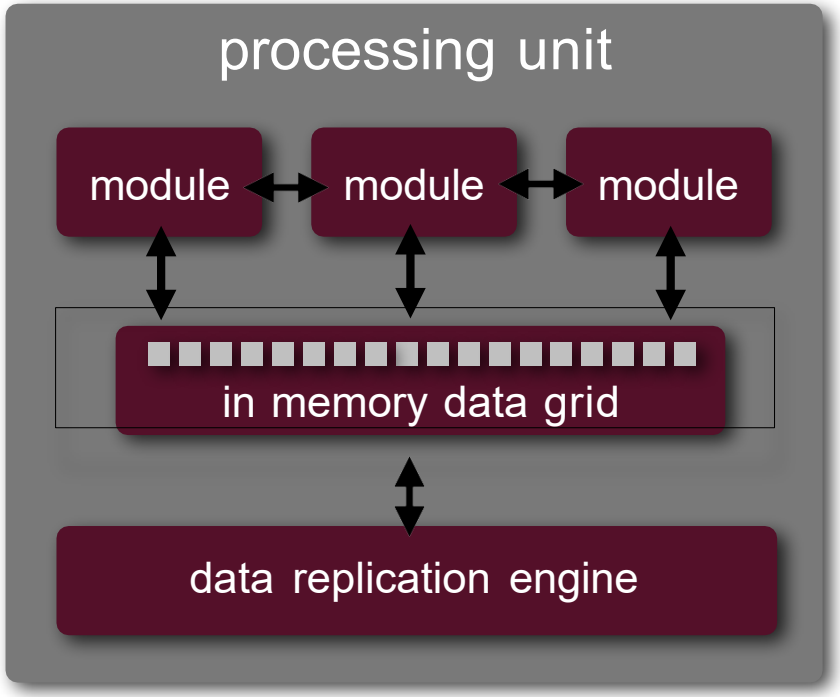
space-based architecture

space-based architecture



space-based architecture

processing unit



space-based architecture

middleware

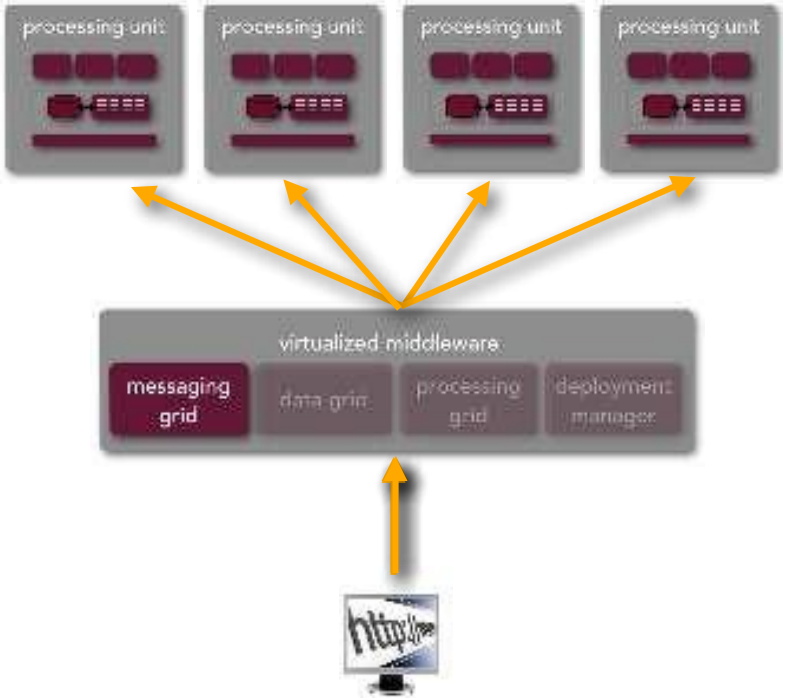
messaging grid

data grid

processing grid

deployment manager

manages input request and session



space-based architecture

messaging grid

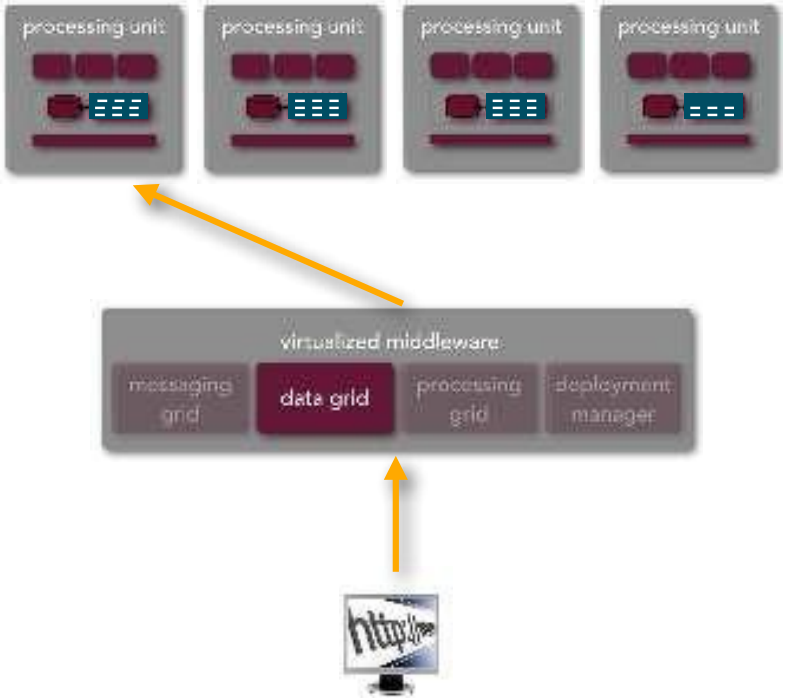
data grid

processing grid

deployment manager

middleware

manages data replication between processing units



space-based architecture

messaging grid

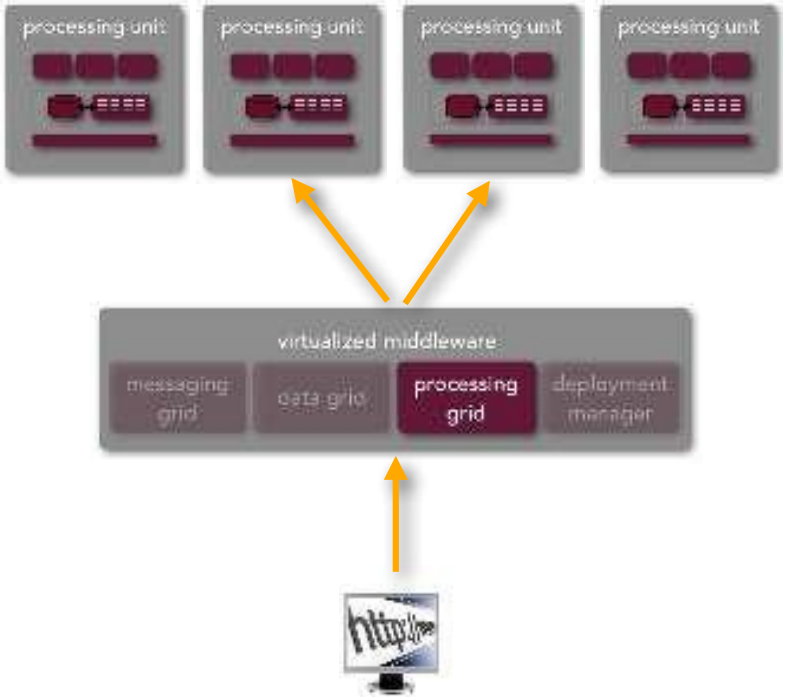
data grid

processing grid

deployment manager

middleware

manages distributed request processing



space-based architecture

messaging grid

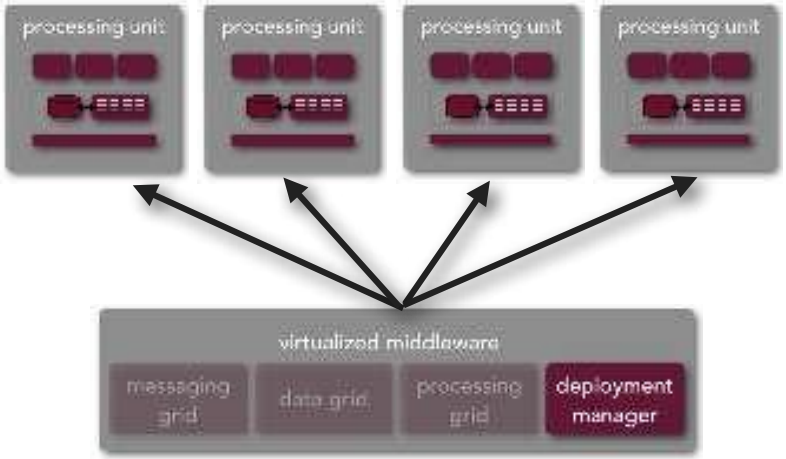
data grid

processing grid

deployment manager

middleware

manages dynamic processing unit deployment



architecture pattern roadmap

layered architecture

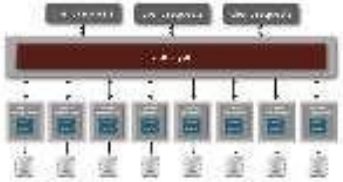
microkernel architecture

event-driven architecture

pipeline architecture

service-based architecture

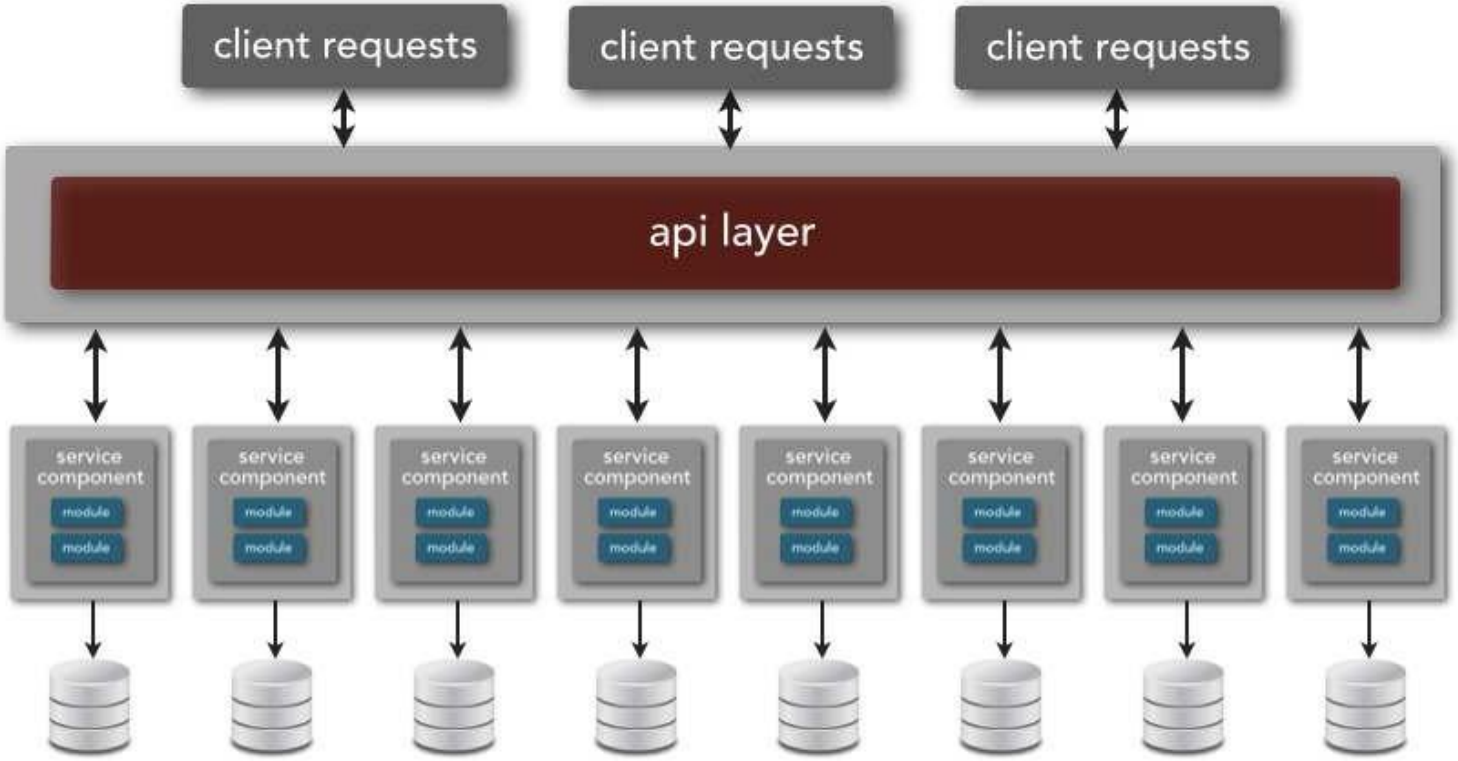
service-oriented architecture



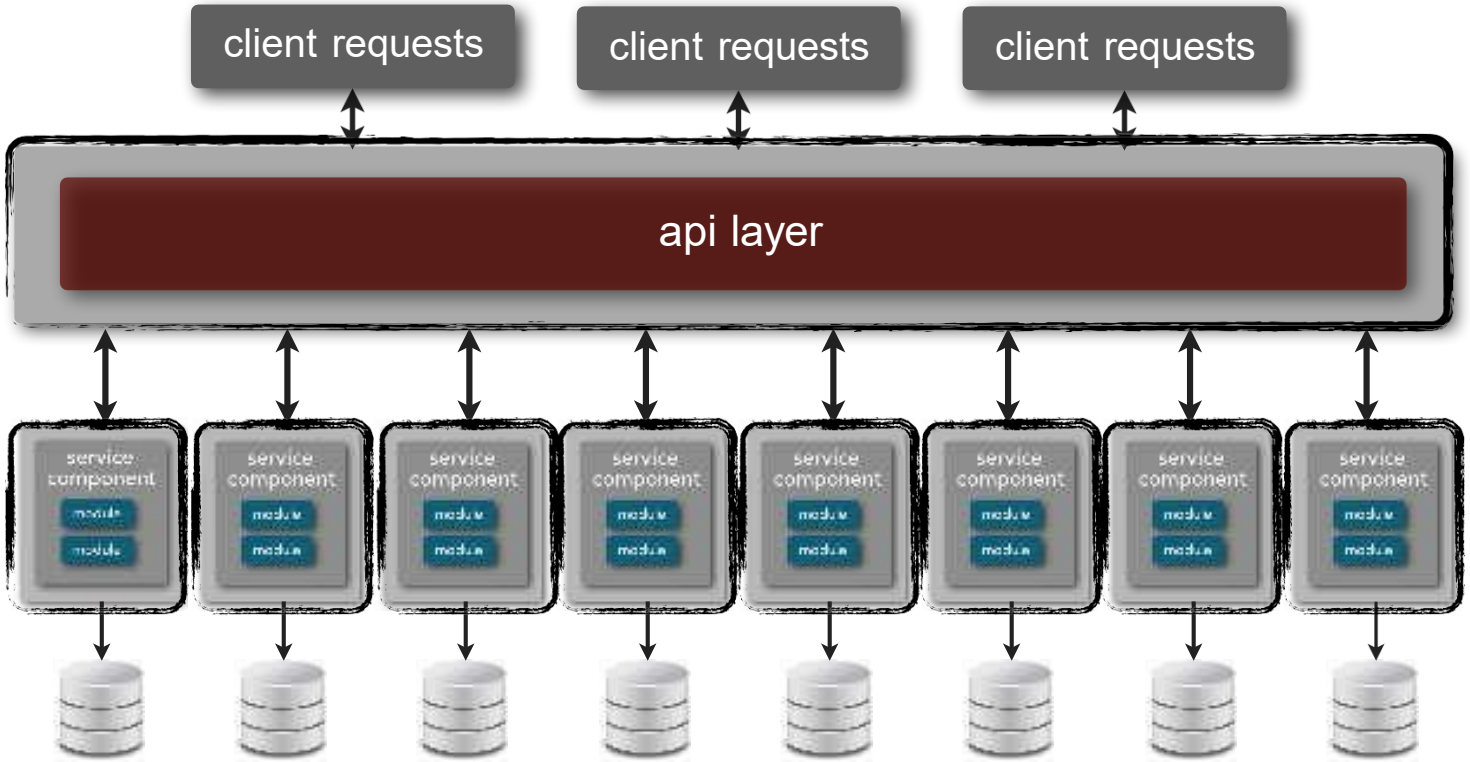
microservices architecture

space-based architecture

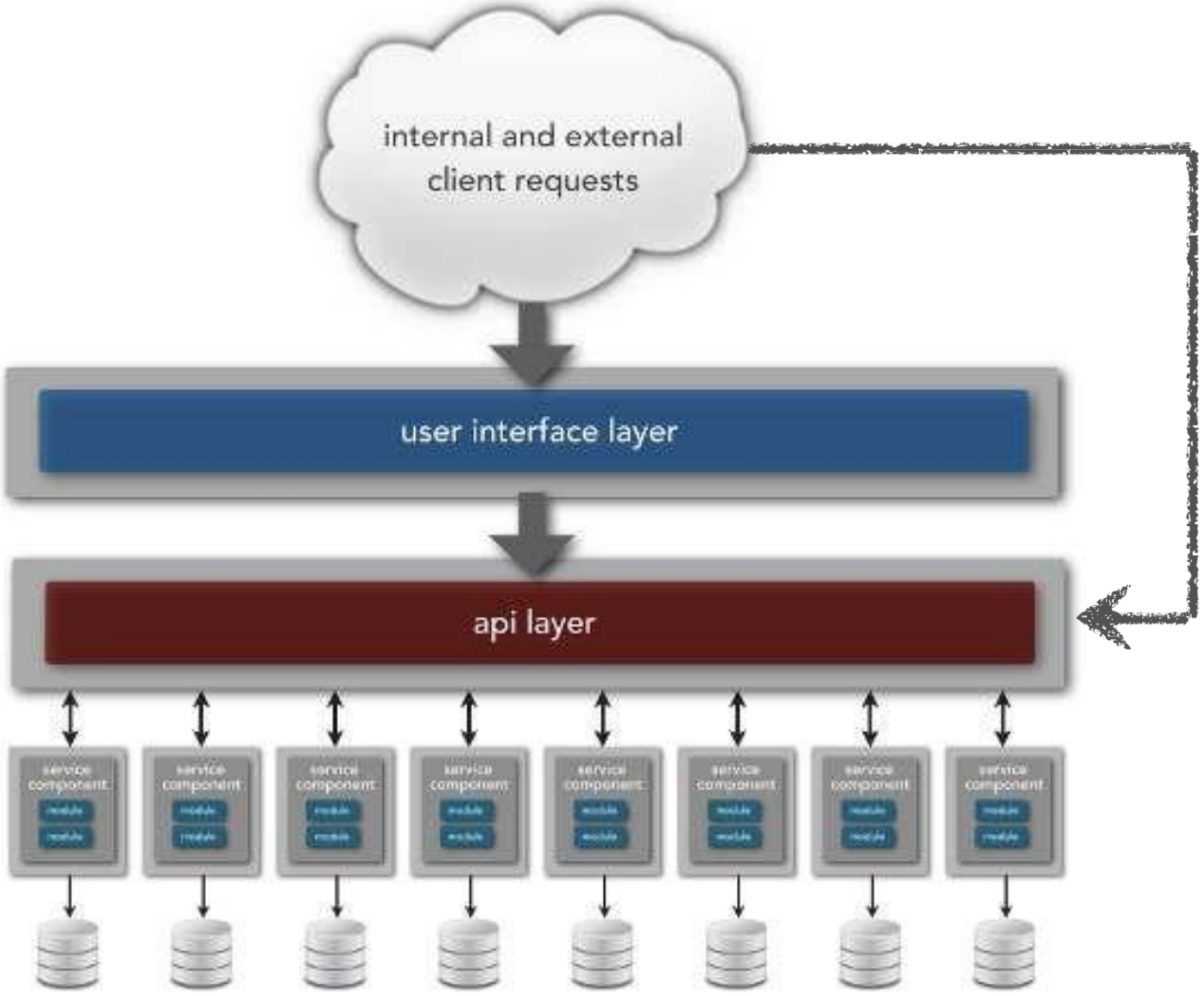
microservices architecture



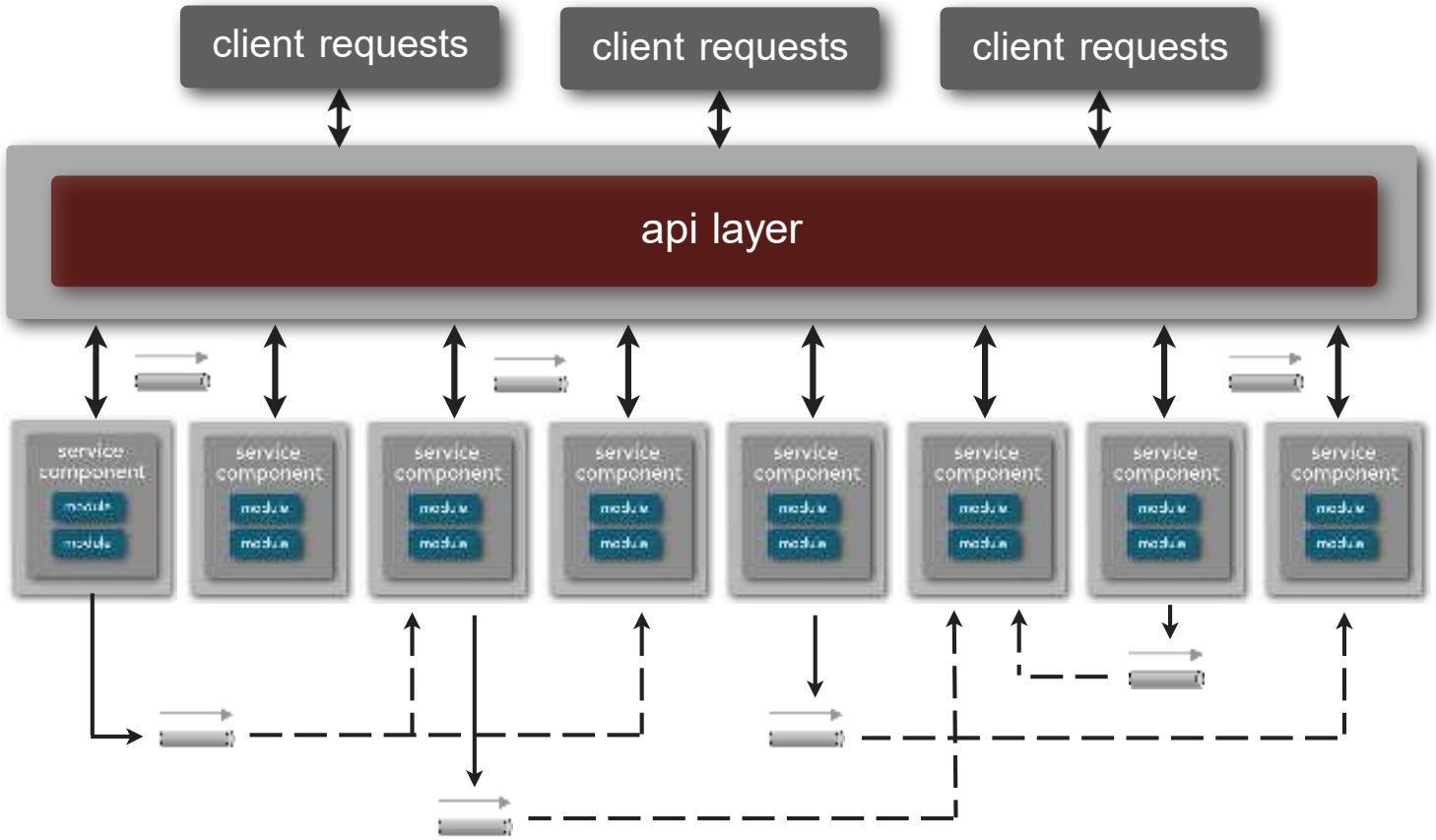
microservices architecture



microservices architecture



microservices architecture



architecture pattern roadmap

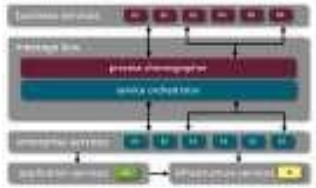
layered architecture

microkernel architecture

event-driven architecture

pipeline architecture

service-based architecture

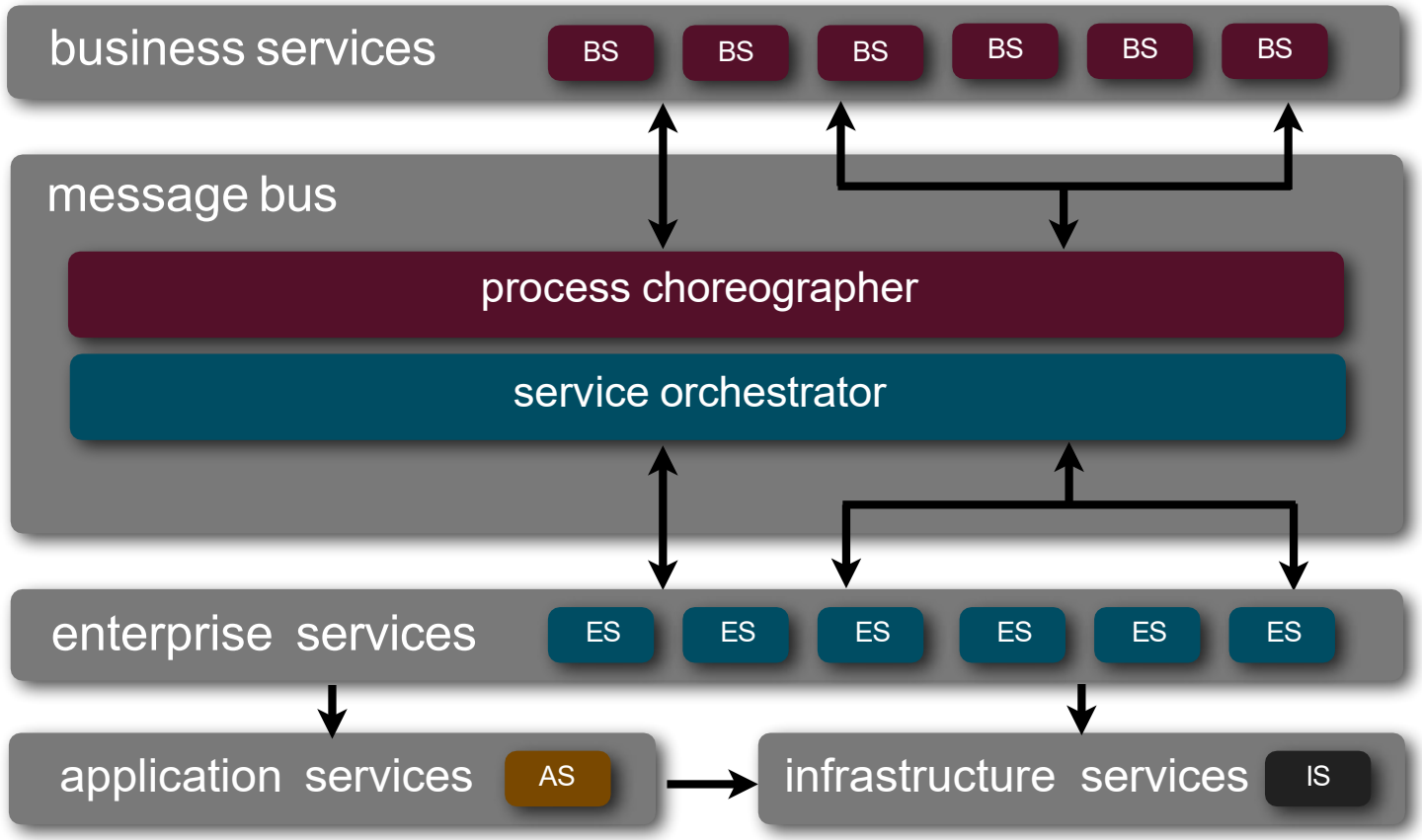


service-oriented architecture

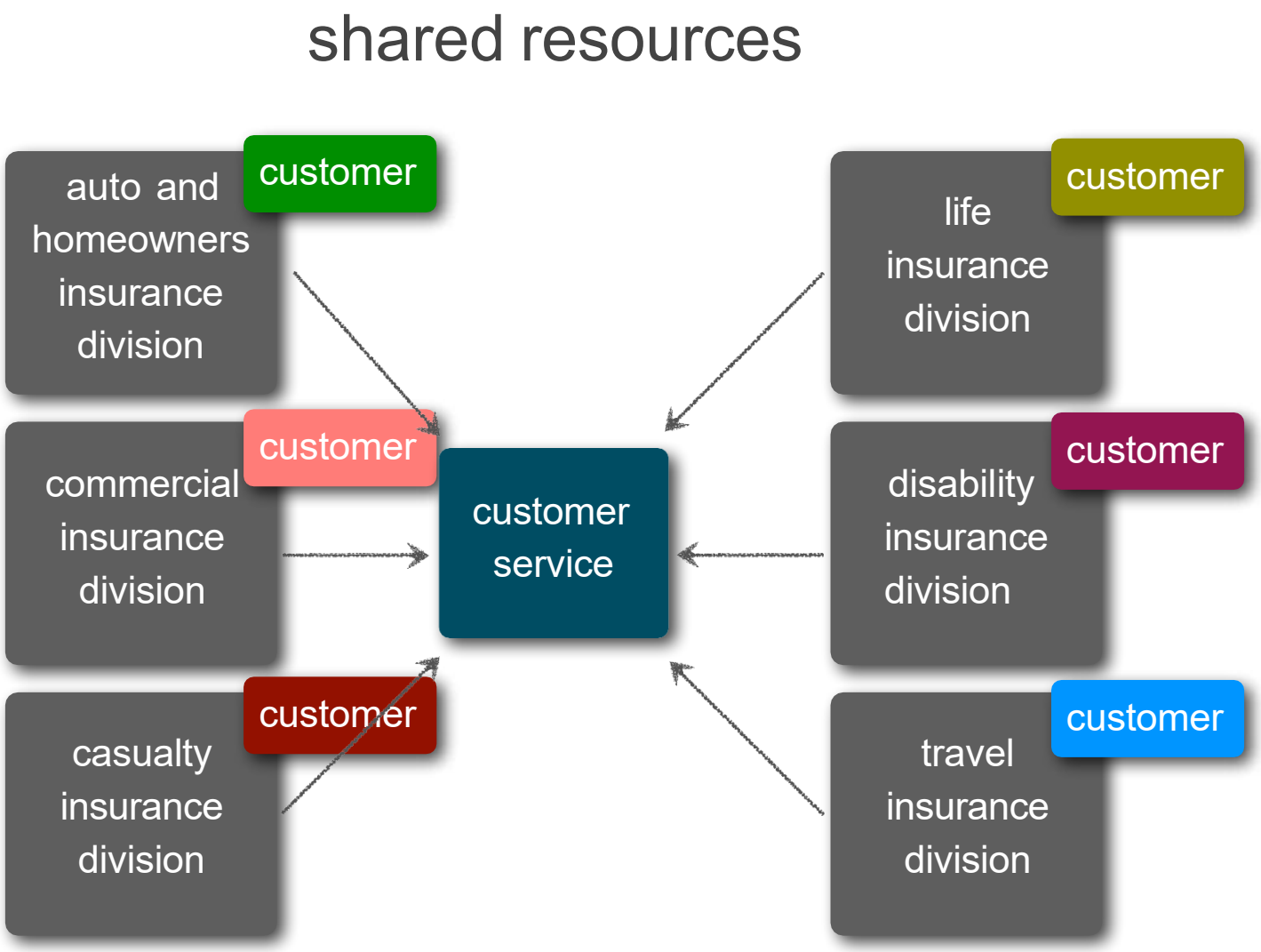
microservices architecture

space-based architecture

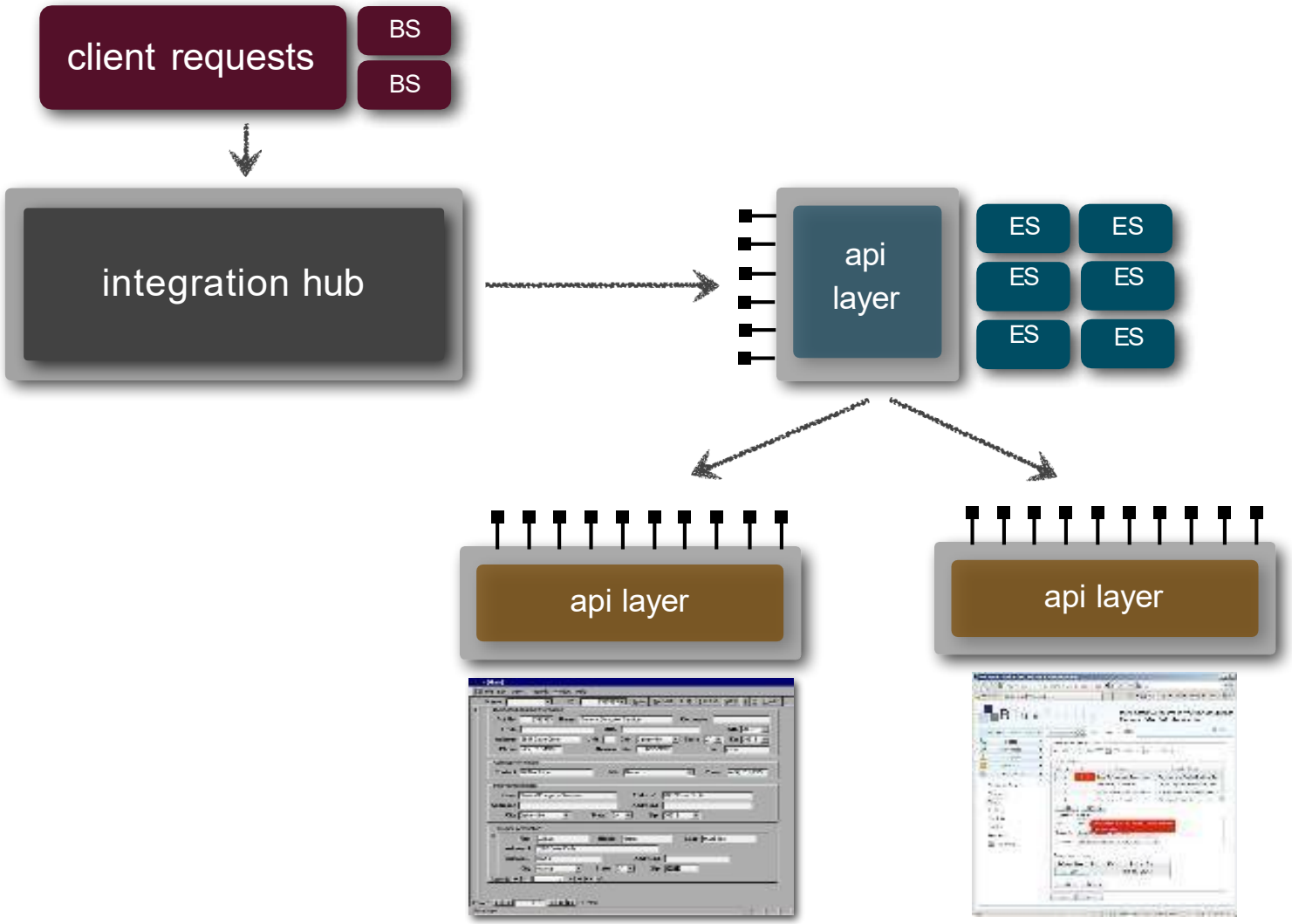
service-oriented architecture



service-oriented architecture



service-oriented architecture



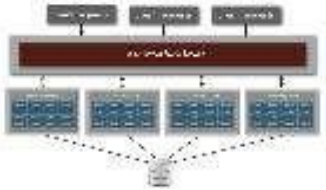
architecture pattern roadmap

layered architecture

microkernel architecture

event-driven architecture

pipeline architecture



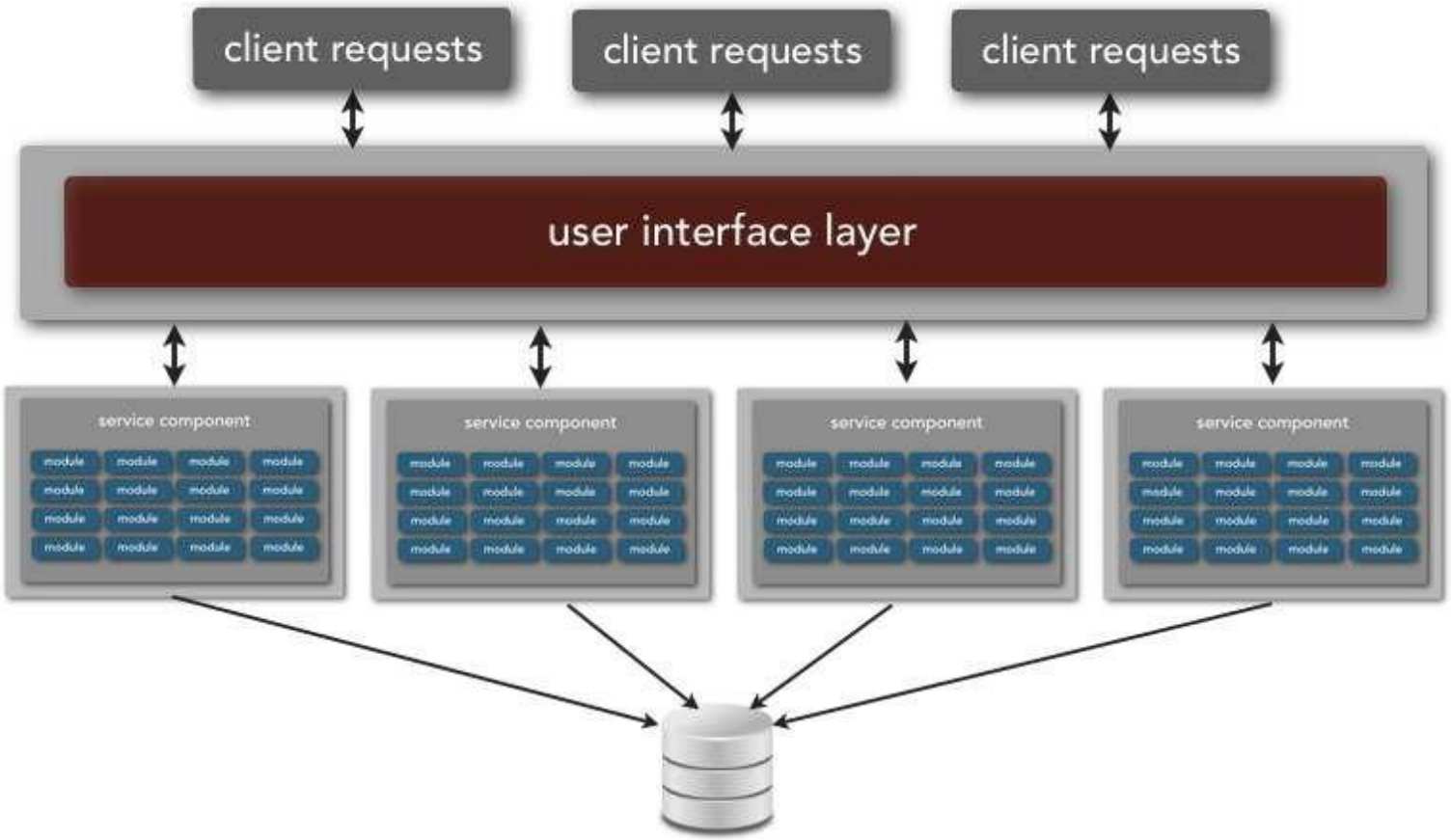
service-based architecture

service-oriented architecture

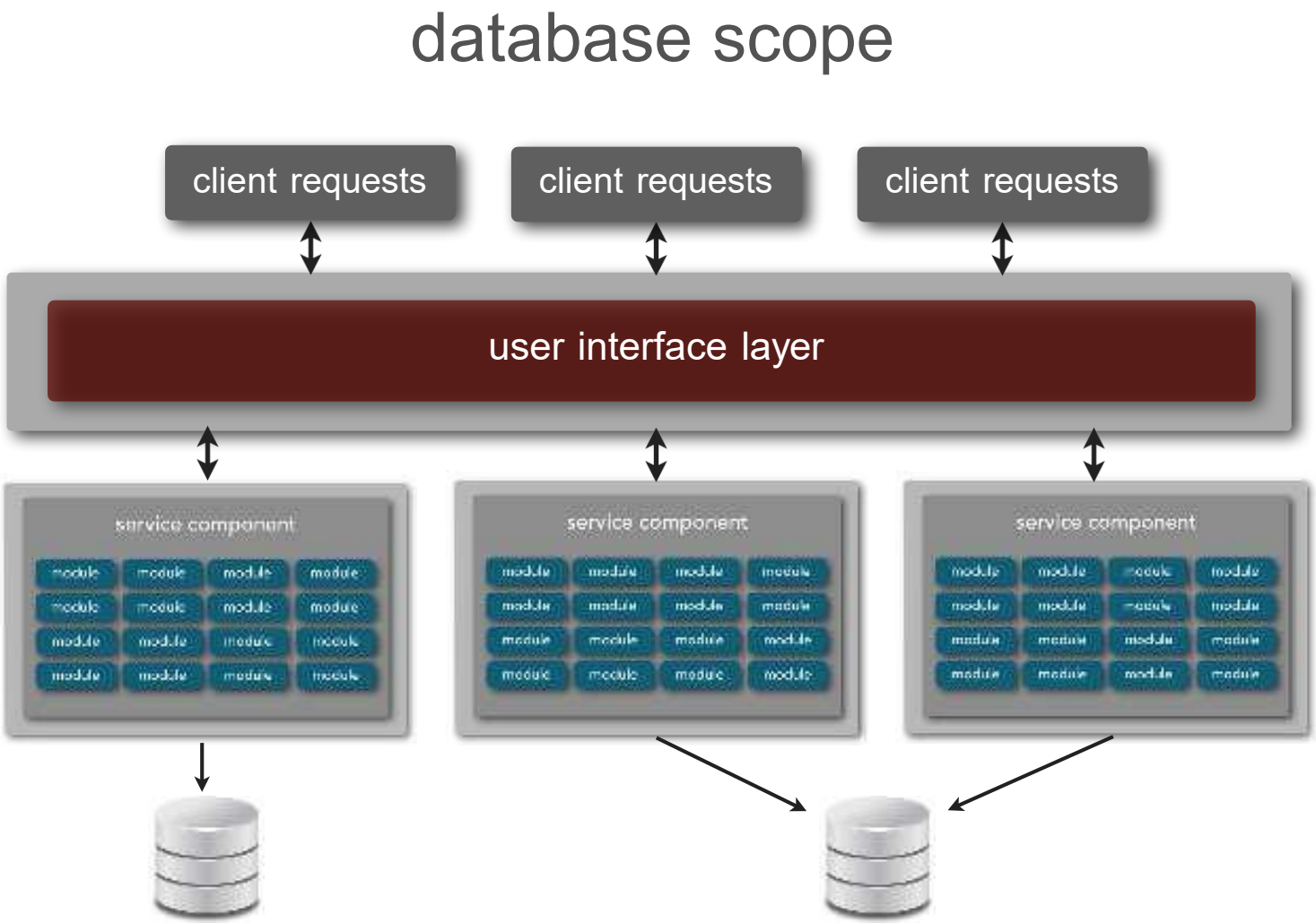
microservices architecture

space-based architecture

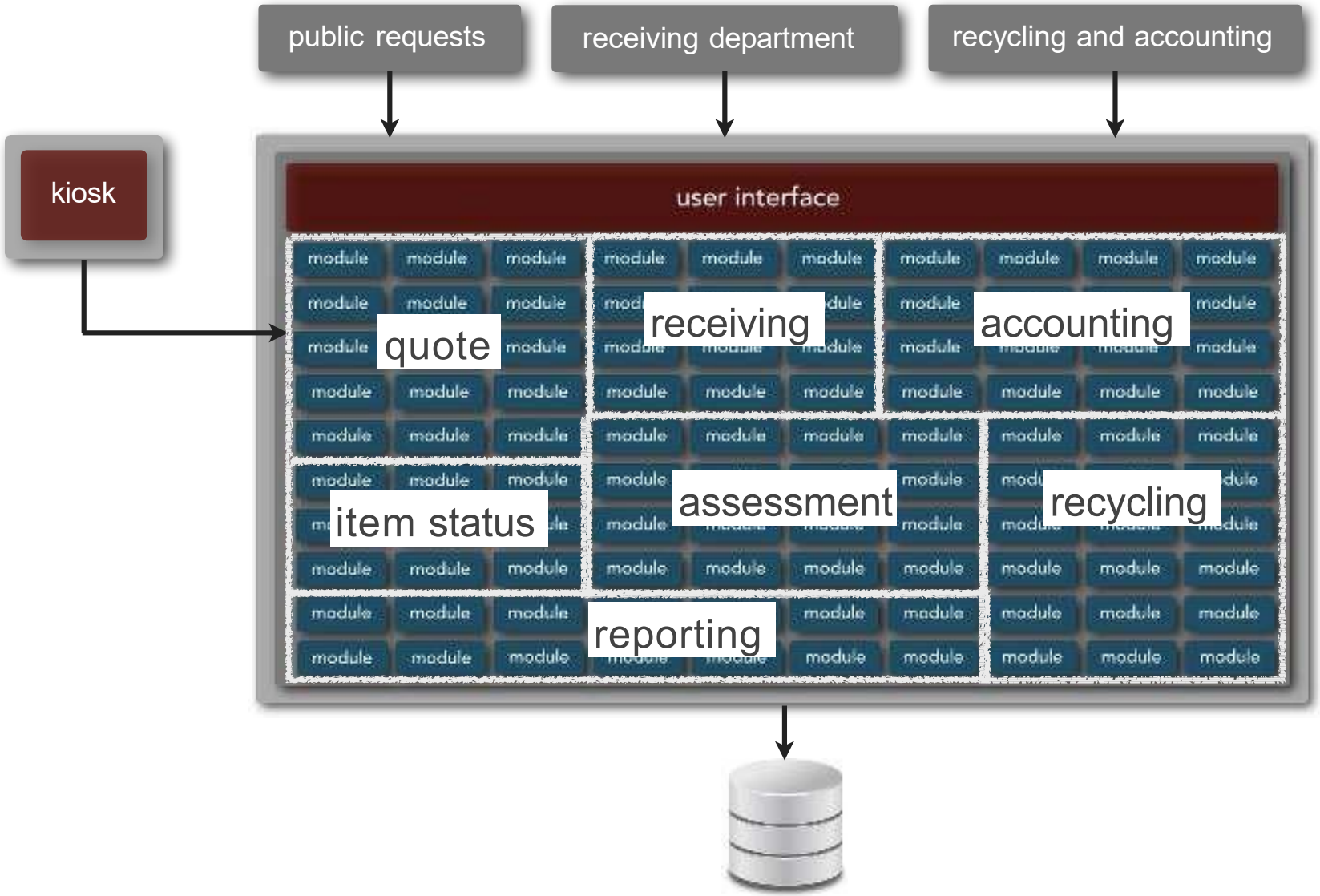
service-based architecture



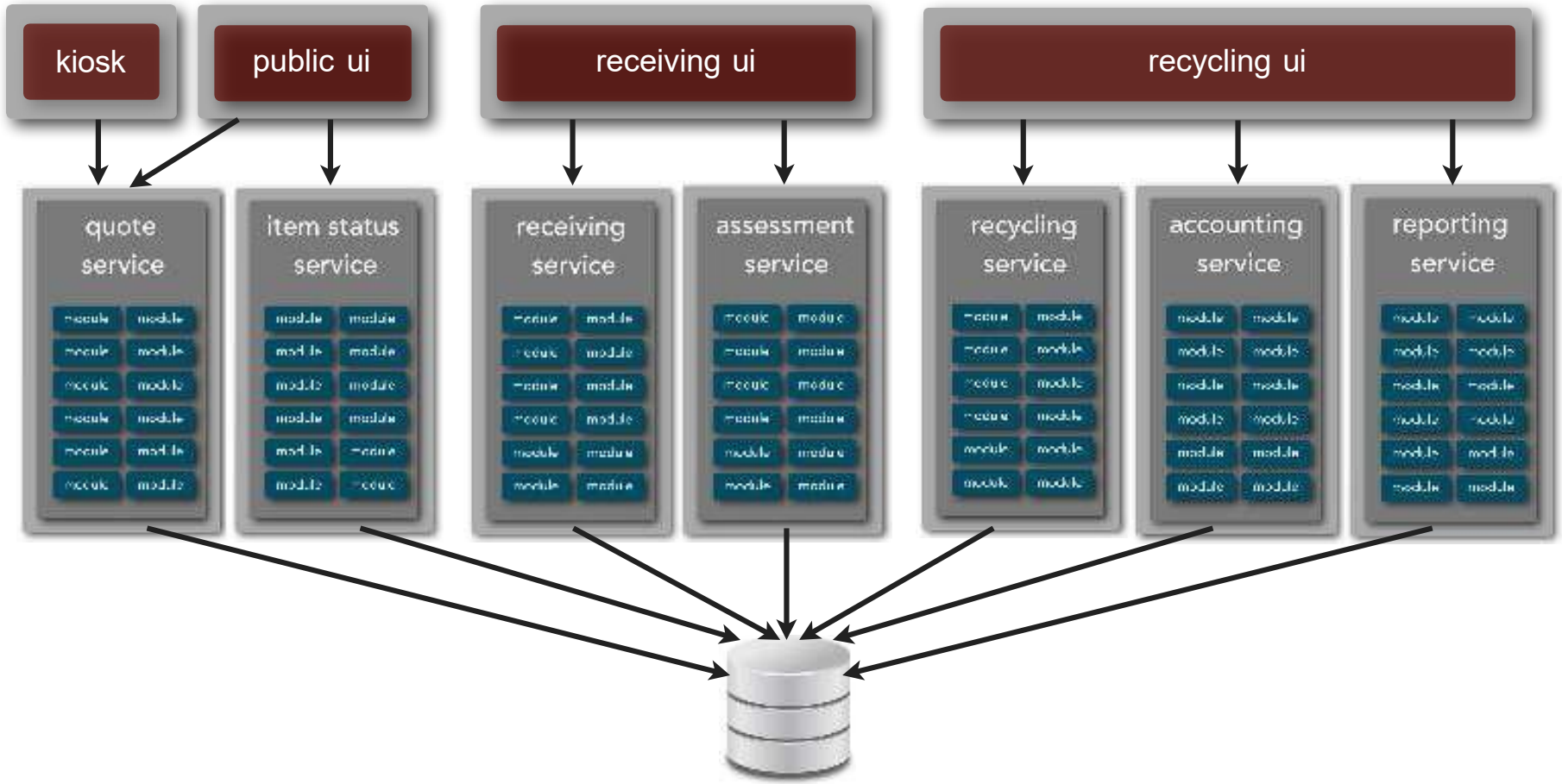
service-based architecture



electronics recycling application

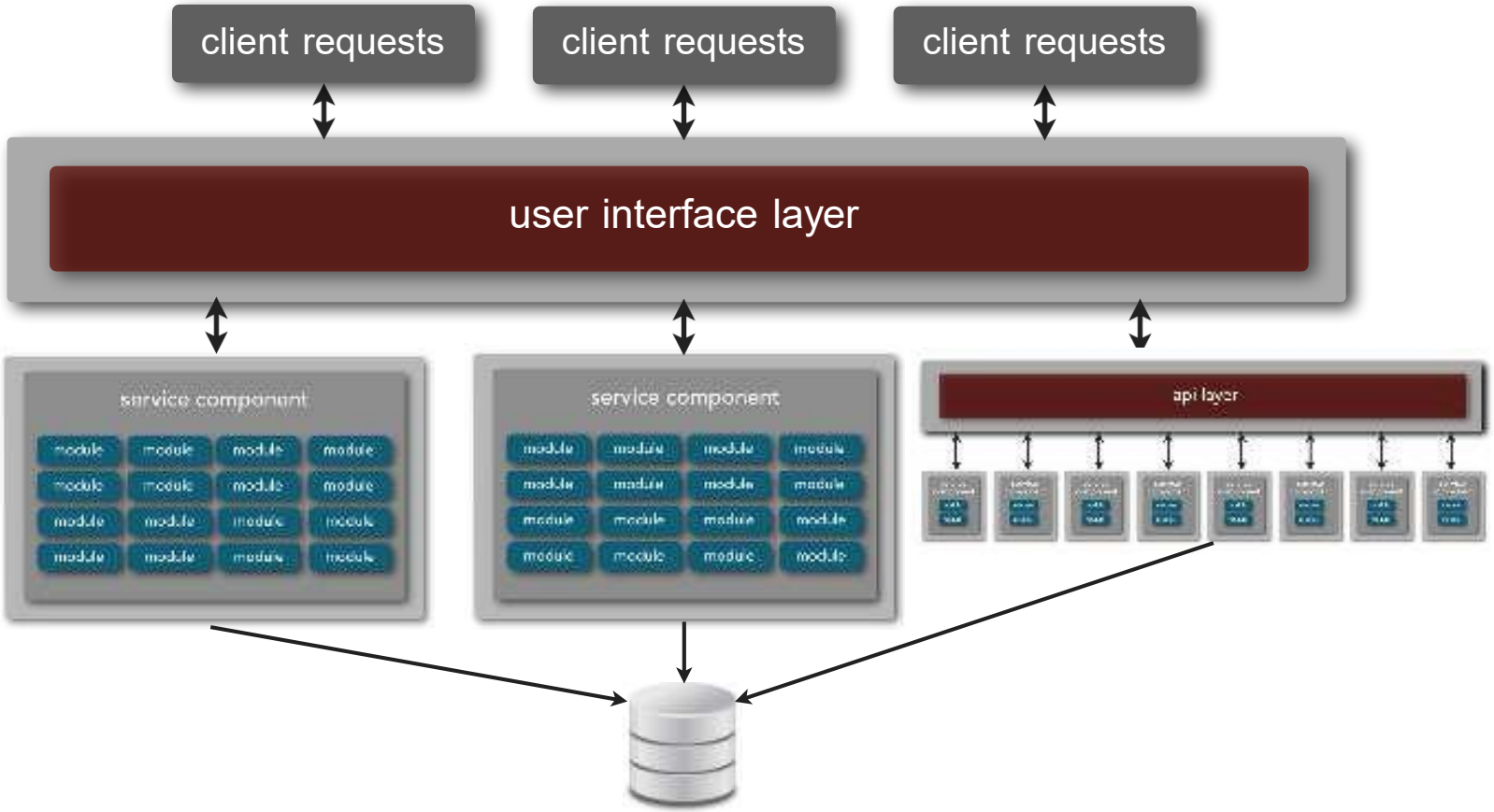


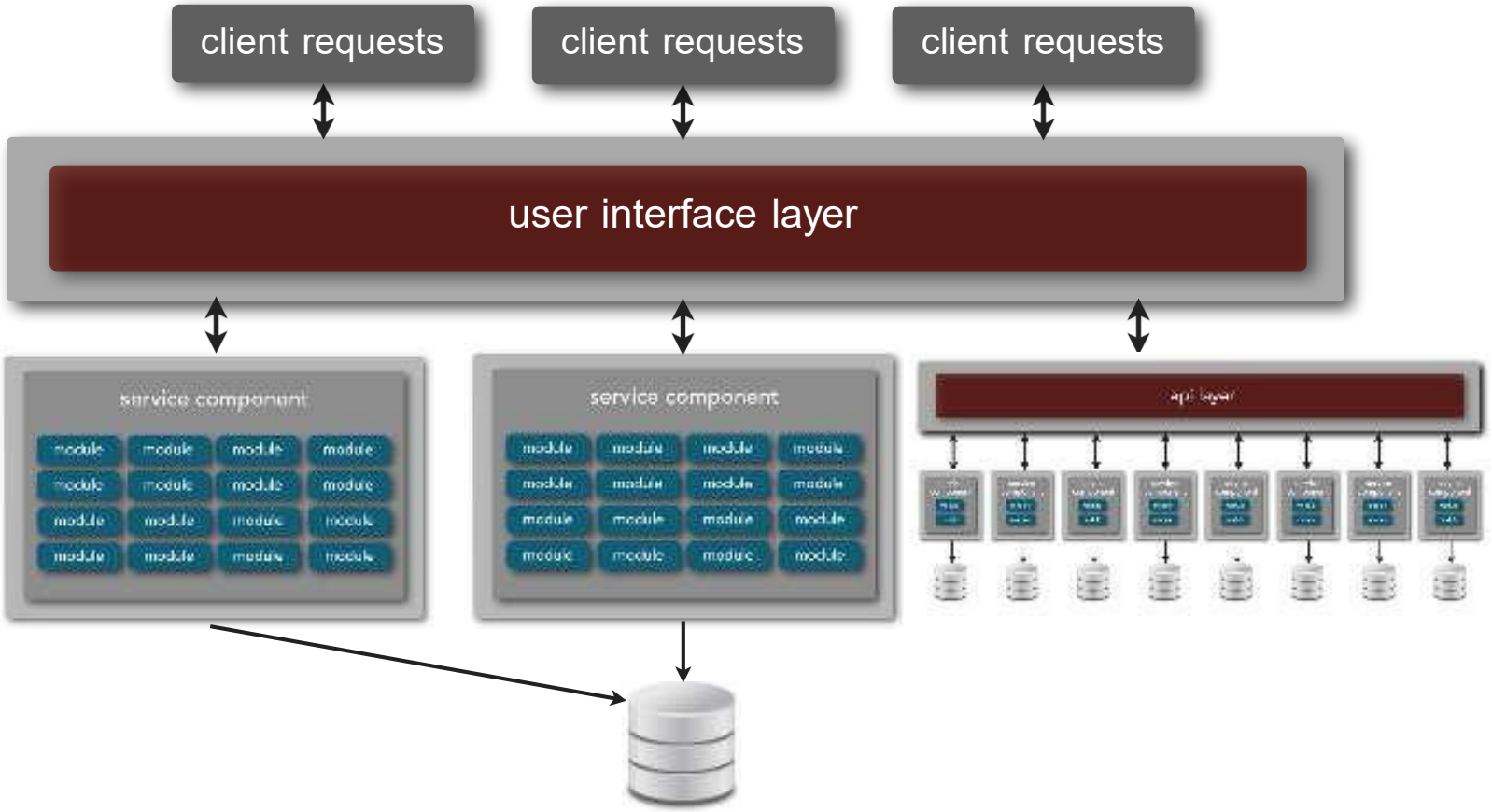
electronics recycling application



service-based architecture

adding microservices



service-based architecture

adding microservices